# Aryan Achary

B.Tech Computer Science and Engineering (second year), Manipal Institute of Technology, India
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**Interests:** 3D computer vision, 3D reconstruction, trustworthy and privacy-aware AI.

## Education

| Qualification | Institution | Year | Score |
|---|---|---|---|
| B.Tech (CSE) | Manipal Institute of Technology, MAHE | 2025–2029 (exp.) | CGPA 8.49 (2 sem.) |
| Class XII (CBSE) | Dalmia Vidya Mandir, Rajgangpur | 2025 | 92% |
| Class X (CBSE) | Dalmia Vidya Mandir, Rajgangpur | 2023 | 97.8% |

## Research

**Auditing private-detail removal in 3D Gaussian Splatting** Oct 2026
*Research proposal, AAAI-27 Undergraduate Consortium (submitted, decision pending)*

- Question: if a private detail is edited out of a 3DGS scene so that no render shows it, can it still be recovered from the Gaussian parameters? Edits are judged against a model retrained without the detail (the unlearning reference).
- Synthetic pilot with a random 256-bit pattern (9 runs per method): recolouring and fine-tuning made renders unreadable, but the pattern stayed readable from the parameters (bit accuracy 0.85–0.96 vs. 0.44–0.58 for the retrained model). Contribution-based selection, as used by common 2D-to-3D segmentation methods, missed about 3% of the Gaussians, which still carried it; rebuilding the whole 3D region reached the retrained range.
- Real phone capture of a desk, with the pattern inserted digitally onto a real blank sheet (2 seeds $\times$ 3 patterns): fine-tuning and the standard densify and opacity-reset clean-up left the pattern readable from per-cell Gaussian counts (0.84–1.00); only rebuilding the whole region stayed in the retrained model's range on all read-outs inside the region.

**Object-landmark correction for dense 3DGS SLAM** 2026
*Equal-contribution co-author with Om Kumar; manuscript in preparation*

- My part: lifting 2D object masks onto the Gaussian map, cleaning them with depth, visibility and multi-view checks, and the evaluation.
- On a dense 119-frame sequence, the cleaned landmark passed the correction guard in 20 of 24 trials (raw lift: 0 of 24) and cut trajectory error from 0.319 m to 0.144 m. With sparser frame sampling the correction raised error by 2–24%; the cause is still open.

## Projects

**Training-free distractor removal for 3D Gaussian Splatting** 2026
*Independent project*

- Cleans an already-trained scene without retraining: SAM 2 propagates text-prompted masks through the capture, and each Gaussian is kept or pruned by a multi-view vote on where its centre projects, with a per-scene threshold.
- On 12 Nerfbusters scenes: 46–94% smaller files and higher held-out PSNR on 11 (largest gains 13.7 $\rightarrow$ 19.2 dB and 15.8 $\rightarrow$ 20.2 dB). The cleanup step takes 1.5–17 s and under 350 MB of memory per scene on a laptop.

**Single-image height estimation from aerial imagery** 2026
*Smart India Hackathon 2026, ISRO/SAC problem statement, team project | github.com/Aryan241/Heimdall*

- Trained a 1.72M-parameter decoder on top of a frozen Depth Anything V2 that predicts per-pixel scale and shift, turning relative depth into metric height above ground (16k GAMUS and Dutch LiDAR tiles). Chose the backbone by its correlation with LiDAR; a metric-depth model scored r = 0.06 on top-down imagery.
- On 8 held-out Dutch LiDAR sites: 5.32 m height RMSE (r = 0.67) vs. 8.37 m for predicting flat ground.

## Leadership

- **Project MANAS**, student AI and robotics team, MIT Manipal: mentor 10+ junior members in machine learning and deep learning; run paper discussions and set coding exercises.
- **e-Yantra** robotics competition (IIT Bombay): team lead of a four-person team.

## Skills

**Languages:** Python, C++, Java. **Libraries and tools:** PyTorch, gsplat, OpenSplat, COLMAP, SAM 2, Depth Anything, Transformers, OpenCV, rasterio, Git, Linux.